

SECTION 3 - MAGNET DELIVERY

3-1 IN-TRANSIT HANDLING REQUIREMENTS



Do not forklift magnet or magnet on pallet.

TABLE 3-1
LIFTING / HANDLING INFORMATION

Magnet	Maximum Weight	Maximum Allowable Tilt
GEMSO 3.0T/940	Magnet: 28,600 pounds (13,000 kg)	30 degrees
	Magnet and Pallet: 29,700 pounds (13,500 kg)	



The magnet CANNOT be shipped by train as serious damage to the magnet's internal suspension system may occur due to the vibration loads introduced by rail systems.

TABLE 3-2
SHIPPING INFORMATION

Magnet	Allowable Shipping Modes	Shipping Capability	Comments
GEMSO 3.0T/940	Airplane, trailer (air ride only), boat or ocean-going ship	Warm	Only tie down to tie down handles and magnet feet.

TABLE 3-3
MAXIMUM LOAD INFORMATION

Magnet	Maximum Longitudinal Load	Maximum Lateral Load	Maximum Vertical Load	Maximum Shock Loads
GEMSO 3.0T/940	5 G	5 G	5 G	None Allowed

Note

Only tie downs to the tie down handles and magnet feet are permitted. See Illustrations 3-1 and 3-2.

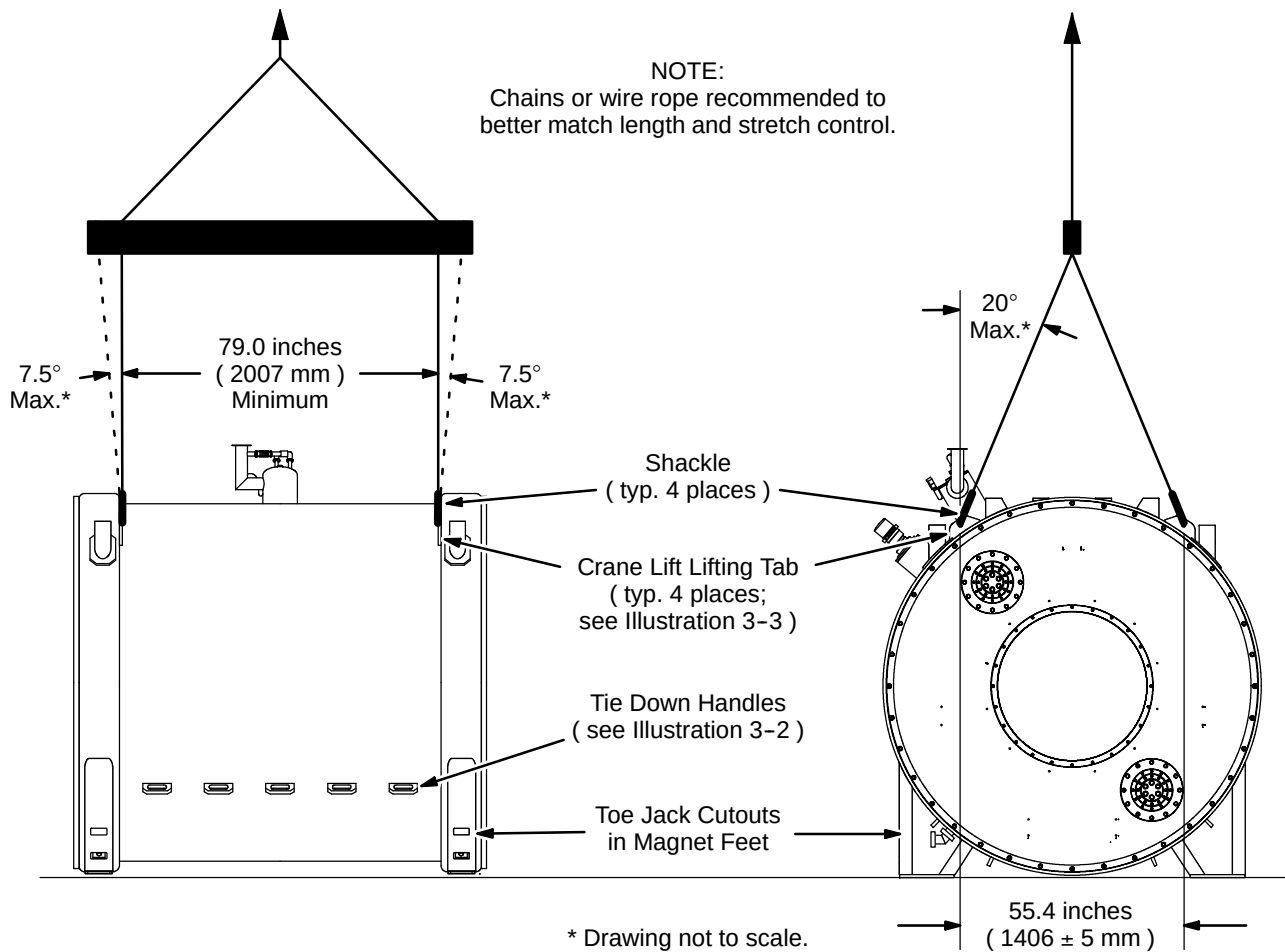
3-2 EQUIPMENT / TOOLS FOR MAGNET INSTALLATION

3-2-1 Lifting / Handling Equipment

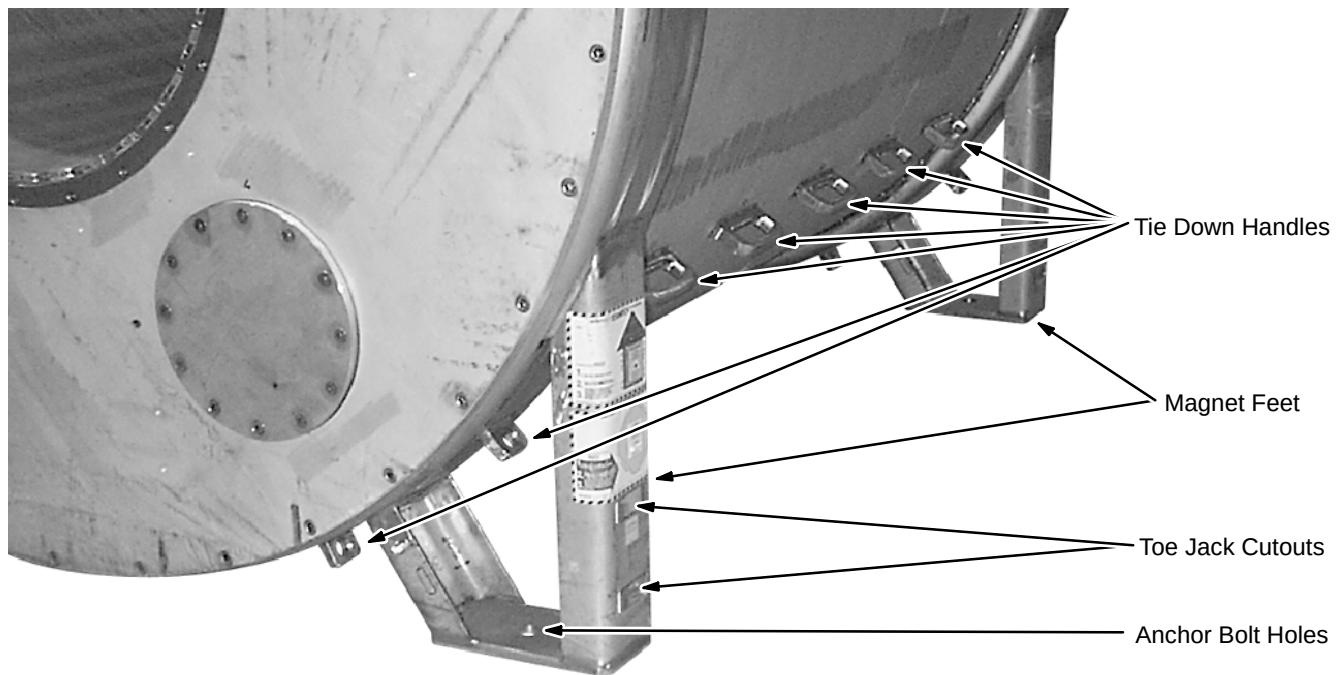
TABLE 3-4
MAGNET LIFTING / HANDLING EQUIPMENT REQUIREMENTS

Item	Qty.	Specifications		Furnished By	Intended Use
Crane Lift					
Crane	1	Total Load:	Magnet: 28,600 pounds (13,000 kg)	Rigger	Unloading or moving magnet
			Magnet and Pallet: 29,700 pounds (13,500 kg)		
Spreader Beam	1	Total Load:	Magnet: 28,600 pounds (13,000 kg)	Rigger	Unloading or moving magnet
			Magnet and Pallet: 29,700 pounds (13,500 kg)		
		Distance Between Lift Points:	79.0 inches (2,007 mm)	Rigger	Unloading or moving magnet
Two Legged Cable Bridle	2	Total Load:	Magnet: 28,600 pounds (13,000 kg)	Rigger	Unloading or moving magnet
			Magnet and Pallet: 29,700 pounds (13,500 kg)		
		Maximum Angle From Vertical:	Side View: 7.5 Degrees End View: 20 Degrees	Rigger	Unloading or moving magnet
Shackle	4	Pin Diameter:	1.18 inches (30 mm)	Rigger	Unloading or moving magnet
		Minimum Work Load Limit:	20,000 pounds (9,100 kg) per shackle		

3-2-1 Lifting / Handling Equipment (continued)

SPREADER BEAM AND BRIDLE SLING
ILLUSTRATION 3-1

3-2-1 Lifting / Handling Equipment (continued)



TIE DOWN AND TOE JACK LOCATIONS
ILLUSTRATION 3-2

3-2-2 Miscellaneous Equipment / Tools

TABLE 3-5
MISCELLANEOUS EQUIPMENT / TOOLS

Item	Qty.	Equipment / Tool Requirement	Responsible	Function
Side-Lifting Toe Jack	4	Must support one end of magnet on two jacks or both ends of magnet on 4 jacks Total weight: 28,600 pounds (13,000 kg)	Rigger	Raising magnet for roller dollies or leveling plates
Roller Dollies	4	Total weight for all dollies combined: 28,600 pounds (13,000 kg)	Rigger	Moving magnet to magnet room
Socket Wrench and Sockets	1	3/4 inch socket (add sockets for crate removal)	Rigger	Removing 1/2 inch crate lag screws
Steel or Aluminum Plate	6	18 inch x 6 foot, .25 inch thick (500 x 2,000 mm, 6.5 mm thick)	Rigger	Moving magnet to magnet room
Pry-Bar	2	Length as required to move magnet	Rigger	Moving / positioning magnet in magnet room
Customer Interface Drawing	1	N / A	GE / ISS	AHZ312475 Identifying magnet dimensions & features
Bill of Material	1	N / A	GE	Checking shipment and aid in magnet installation
Level	1	36 inch (1000 mm) length	Rigger	Leveling magnet
M10 Screws	4	Minimum length: 40 mm	Rigger	Supporting level during side-to-side magnet leveling
Magnet Leveling Kit	1	N / A	GE / SYS	46-260888G4 Leveling magnet
Hammer and Wooden Driving Block	1		Rigger	Leveling magnet
Magnet Boltdown / Seismic Kit	1	Apply sufficient torque to provide 7,000 pounds (3,175 Kg) clamping force per anchor. Torque requirements to be supplied by screen room vendor.	Screen Room Vendor	Bolting down magnet
Torque Wrench and Socket	1	Torque range: 58 to 258 ft-lbs. Deep socket, size dependant on anchor selection	Rigger	Bolting down magnet

3-3 UNLOADING MAGNET

CAUTION

The lifting hooks of the spreader beam **MUST** be located as shown in Illustration 3-1. The magnet can be lifted by a crane which meets the specifications stated in Table 3-4. Failure to follow these requirements can result in serious damage to the magnet.

1. Position the hook of a crane meeting the specifications stated in Table 3-4 centrally over the magnet to ensure a vertical lifting force on the lifting cables. See Illustration 3-1.

WARNING!

DO NOT USE THE TIE DOWN HANDLES SHOWN IN ILLUSTRATIONS 3-1 AND 3-2 TO LIFT THE MAGNET AS THEY WILL BREAK AND DAMAGE THE MAGNET.

2. Attach the rigging to the lifting tabs at each end of the magnet. See Illustrations 3-3.



CRANE LIFT LIFTING LUG
ILLUSTRATION 3-3

3-3 UNLOADING MAGNET (continued)



Impacts / jolts to the magnet while lifting / moving / lowering the magnet can cause expensive internal magnet damage. Lift / move / lower smoothly. Do not allow the magnet to bump or hit anything forcefully.

3. Lift the magnet with the crane.
4. Smoothly move the magnet to the desired location and carefully lower the magnet to rest on a flat surface.

3-4 PALLET REMOVAL

1. Cut the protective outer plastic bag / wrap loose from the shipping pallet but leave the bag / wrap around the magnet.

Note

Leave the protective plastic bag / wrap around the magnet as long as practical, especially at an unfinished site.

2. Loosen the bolts in each leg securing the magnet to the pallet.



Impacts / jolts to the magnet while lifting / moving / lowering the magnet can cause expensive internal magnet damage. Lift / move / lower smoothly. Do not allow the magnet to bump or hit anything forcefully.

3. Lift the magnet clear of the pallet using a crane. See Section 3-3, Unloading Magnet.
4. Remove the pallet.
5. Carefully lower the magnet to rest on a flat surface.

3-5 PROTECTIVE PACKAGING REMOVAL

Note

Leave the protective plastic bag / wrap around the magnet as long as practical, especially at an unfinished site.

CAUTION

Care must be taken not to scrape or hit the sides of the magnet.

The magnet is shipped wrapped in a protective plastic bag / wrap. Inside the bag the Service Turret and Shield Cooler Coldhead are surrounded by additional protection. Care must be taken in removing the plastic bag to avoid damage to various magnet components. Leave the protective boxes / wrap around the Service Turret and Cryo-cooler Coldhead until the magnet is anchored in the magnet room.

1. Remove the protective outer plastic bag / wrap. See Illustration 3-4. Leave the protective boxes / wrap around the Service Turret and Shield Cooler Coldhead until the magnet is anchored to the magnet room floor.



MAGNET AS SHIPPED (DOMESTIC)
ILLUSTRATION 3-4

SECTION 4

MAGNET SYSTEM COMPONENT CHECK (BY GE SERVICE)

IMPORTANT !!!

It is important to establish if any damage was sustained by the magnet or the system components before unloading from the truck / carrier. Report any damage found in compliance with the "Damage in Transportation" note on the back side of the Service Manual Title Page. Do not unload the magnet until the magnet and its shipping boxes have been checked.

Report any missing components to the person identified on the Magnet Components Bill of Material. Report any damaged components in conformance with "Damage in Transportation," located at the front of this manual.

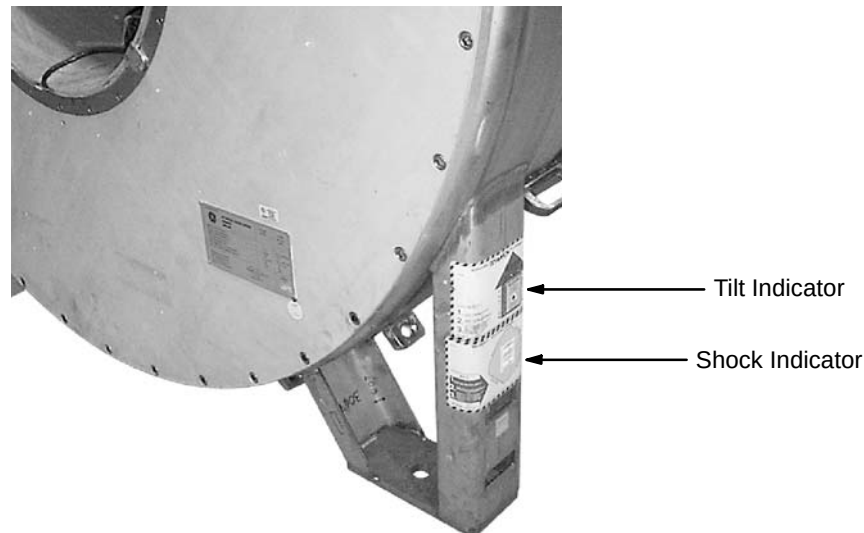
Report all problems found to the Regional Magnet & Cryogenics (MAC) Team Leader.

Note

Shipment of magnet system components to the installation site may occur as a complete system or as a drop shipment of major system components. Verify that all required magnet system components are present at the site to assure a complete and timely installation.

1. Locate the Bill of Material shipped with the magnet. It identifies the boxes shipped with the magnet. Check that all boxes indicated are present.
2. Check the contents of each box against its packing list when the boxes are brought into the MR Site. See Table 4-1.
3. Inspect the magnet and the content of the shipping boxes for physical damage.
4. Check the Shock and Tilt Indicators shown in Illustration 4-1. If they have been tripped, the magnet could be damaged.

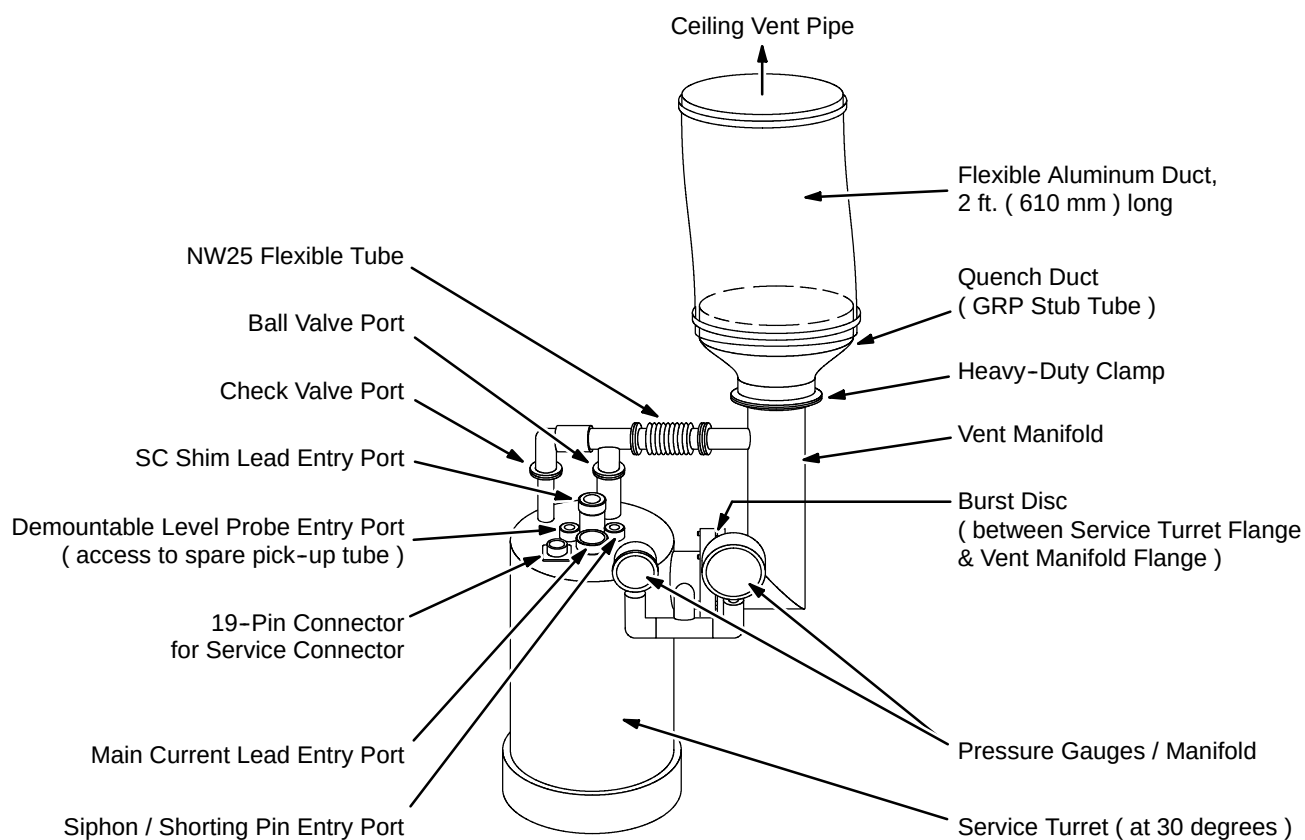
SECTION 4 - MAGNET SYSTEM COMPONENT CHECK (BY GE SERVICE) (continued)



SHOCK & TILT INDICATORS
ILLUSTRATION 4-1

5. Check SC Shim Lead Connector alignment:
 - a. Open the SC Shim Lead Entry Port on top of the Service Turret. See Illustration 4-2.
 - b. Shine a flashlight through the port to visually check the alignment of the SC Shim Lead Connector with the SC Shim Lead Entry Port.
 - c. Close the port.
 - d. Report any obvious misalignment.
6. Check Siphon / Shorting Pin Connector alignment:
 - a. Open the Siphon / Shorting Pin Entry Port on top of the Service Turret. See Illustration 4-2.
 - b. Shine a flashlight through the port to visually check the alignment of the Siphon Fill Cup with the Siphon / Shorting Pin Entry Port.
 - c. Close the port.
 - d. Report any obvious misalignment.
7. Report any missing, damaged / misaligned components or tripped Tilt / Shock Indicators immediately.
 - Report missing components to the person identified on the Magnets Components Bill of Material.
 - Report damaged / misaligned components or tripped Tilt / Shock Indicators in conformance with "Damage in Transportation," located at the front of this manual.

SECTION 4 - MAGNET SYSTEM COMPONENT CHECK (BY GE SERVICE) (continued)

SERVICE TURRET
ILLUSTRATION 4-2

SECTION 4 - MAGNET SYSTEM COMPONENT CHECK (BY GE SERVICE) (continued)

TABLE 4-1
3T940 BILL OF MATERIAL

QUANTITY	GE P/N	MAGNEX P/N	DESCRIPTION
Box 1 - Documentation			
1		MAN11009CV01	Magnex 3T/940 AS Manuals (includes the E5011 Liquid Helium Level Monitor Manual and the E7006 Emergency Discharge Unit Manual)
1	2295400		3T/940 Active Shield Magnet & Cryogenics Subsystem Manual (CD)
1		LOGBOOK	Log Book
1			ATR w/ Final Test Data, 3.5 In. Floppy Disc (Florence shipments)
1			ICD Card for Magnet / Collector
1			ICD Card for Compressor
Box 2 - Safety Signage Kit 46-258770G4			
10	46-255326P1		5 Gauss Exclusion Zone Warning Sign
2	46-255325P1		Security Zone Warning Sign
50	46-255813P1		Magnetic Item Warning Label, 5 x 3 In. (127 x 76 mm)
2	2289812		Warning Sign: Authorized Personnel Only
50	46-255814P1		Magnetic Warning Label, 2.5 x 1.5 In. (64 x 38 mm)
2	46-282214P1		Coolant Caution Label
1	46-301671P1		Warning Placard
Box 3 - De-Transit Kit			
4		DHC412690	CCS Cat Flap
4		DHC412691	GCS Cat Flap
4			Insulation Pads (from unit)
96	2324428	FB19050012	M6 Screws
48	2324795	FS13120025	M12 x 25 mm CSk. Hd. Screw, SS (transit blanking plate)
1 roll	2212419-5		Aluminized Mylar Tape, 1 In. (25 mm) Wide
Box 4 - Exhaust Ducting			
1	2292545	AHC318635	8 In. (203 mm) Quench Duct Assembly
1	2325534	P194000013	8 In. (203 mm) Flexible Flue Duct, 24 In. (610 mm) Long
2	2292512	DHC118913	8 In. (203 mm) Duct Clamp (2 sections; 4 total pieces)
4	46-318764P6	FS10060025	M6 x 25 mm Soc. Hd. Cap Screw, Stl.
8	2109878-3	FS40000006	M6 Flat Washer, Stl.
4	2324385	FS50000006	M6 Nut, SS
1	2293036	P208000051	Heavy-Duty Clamp
1	2293038	P208000053	Gasket

SECTION 4 - MAGNET SYSTEM COMPONENT CHECK (BY GE SERVICE) (continued)

TABLE 4-1 (continued)
3T940 BILL OF MATERIAL

QUANTITY	GE P/N	MAGNEX P/N	DESCRIPTION
Box 5 - Vent and Baffles			
1	2255519	AHC114786	Copper Berillium Baffle Assembly
1	2255520	AHC315883	Shim Lead Baffle Assembly
1	2292514	AHC309787	Elbow Tube Assembly
1	2292513	DHC409800	Bypass Manifold Tube, Short
1	2292541	AHC315227	Tee Manifold Assembly
2	2255516	P200000019	Burst Disc / Burst-Tel
1	2294711	DVU318651	NW25 - NW25 Flex Hose, 110 mm Long
2	46-241180P3	P171022530	O-Ring, 22.5 mm I.D. x 3.0 mm
3	2294083-16	P208000038	KF25 Clamp
3	2294083-17	P208000039	KF25 O-Ring with Center
Box 6 - Cables			
1	2295525-11	C0090085	Cable, EDU to He Monitor, 8.5 M (27.9 Ft.)
1	2295525-10	C0090170	Cable, EDU to He Monitor, 17 M (55.8 Ft.)
1	2295525-9	C0091170	Service Cable
2	2255533	P180000002	Main Power Lead, IEC American 3-Pin
Box 7- Magnet Leveling Kit 2343055			
1	2343055		3T/940 Magnet Leveling Kit
Box 8 - 3T/940 Passive Shim Installation Kit 2292493			
1 bag		FB14040007	M4 x 7 mm Screw, Brass (~ 1000)
1 bag		DRZ415933	2 mm Tapped Alum. Sq. (~ 500)
1 bag		DRZ415932	3 mm Alum. Shim, Sq. (~ 500)
1 bag		DRZ415930	1 mm Tapped Iron Shim, Rect. (~ 500)
1 bag		DRZ415929	2 mm Tapped Iron Shim, Sq. (~ 500)
1 bag		DRZ418785	1 mm Iron Shim, Sq. (~ 500)
1 bag		DRZ415928	3 mm Iron Shim, Sq. (~ 500)
1 bag			2 mm Iron Shim (~ 1000) - Not Required

SECTION 4 - MAGNET SYSTEM COMPONENT CHECK (BY GE SERVICE) (continued)

TABLE 4-1 (continued)
3T940 BILL OF MATERIAL

QUANTITY	GE P/N	MAGNEX P/N	DESCRIPTION
Items Packed Separately			
1	46-317271G1		Oxygen Monitor
1	2295525-8	E7006	E7006 EDU (Magnet Rundown Unit)
1	2255532	E5011	E5011 He Monitor mounted in rack frame:
1		P214000006	E5011 He Monitor
1		P214000044	19 In. Subrack Frame (84F)
1	2255515	P222000019	Ladder
1	2321775	DHC115301	Adjustable Service Platform
1	2255522	AUM117523	Syphonic Shorting Pin & Tool
1	2255856	AUE116445	Helium Level Probe
Pallet 2 - Cryocooler and Compressor			
1	2255524	P196000025	CoolPak 6000 Compressor*
1 box	2255527	P196000008	Semi-Flexible Gas Lines, 11 M (36 Ft.) (2 per box)
1	46-294150P56	P196000030	CP6000 Compressor Power Cable, U.S., 3.5 M (11.5 Ft.)
1	46-294144P2	P176000130	Cable, Compressor to Filter Panel, 50Ft. (15.24 M)
2	2251611		Flexible Corrugated Tubing, Orange
* CoolPak 6000 weight: 280 pounds (130 Kg). Dimensions: 29.5 x 24.0 inches (750 x 610 mm) x 26.8 inches (680 mm) tall.			

SECTION 5 - MOVING MAGNET TO MR SUITE

5-1 INTRODUCTION

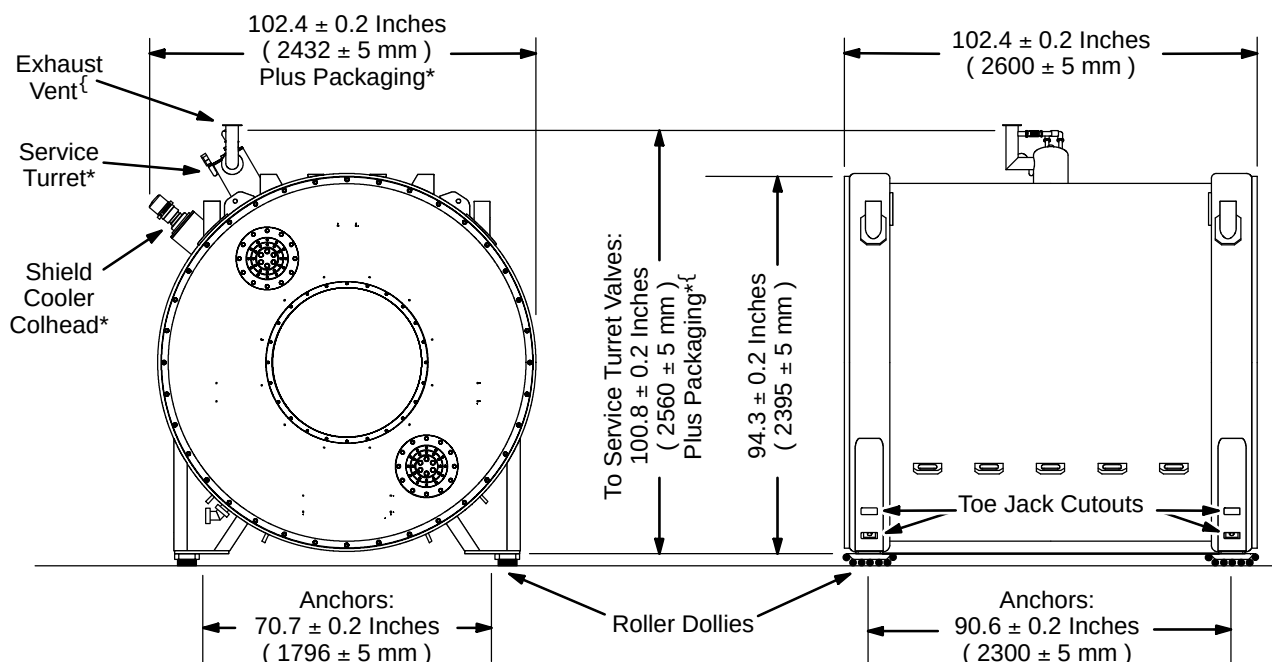
WARNING!

REFER TO SECTION 3, MAGNET DELIVERY, BEFORE MOVING THE MAGNET USING A CRANE.

Once the magnet has been moved to the building using a crane — covered in Section 3, Magnet Delivery — it needs to be moved to the magnet room. Roller dollies in the arrangement shown in Illustrations 5-1 and 5-2 are recommended for moving the magnet inside a building. Place steel floor plates along the magnet delivery route when using roller dollies.

Note

Recommended MINIMUM clearances along movement route: Vertical^{}: 105.5 inches (2680 mm);
Horizontal: 140.75 inches (3575 mm) to rotate (turn) magnet as shown in Illustration 5-3.



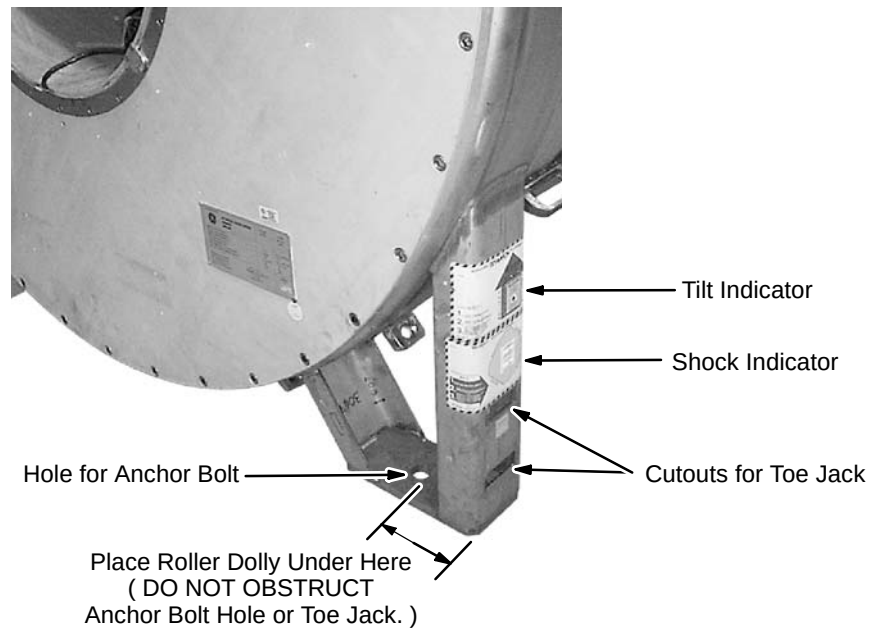
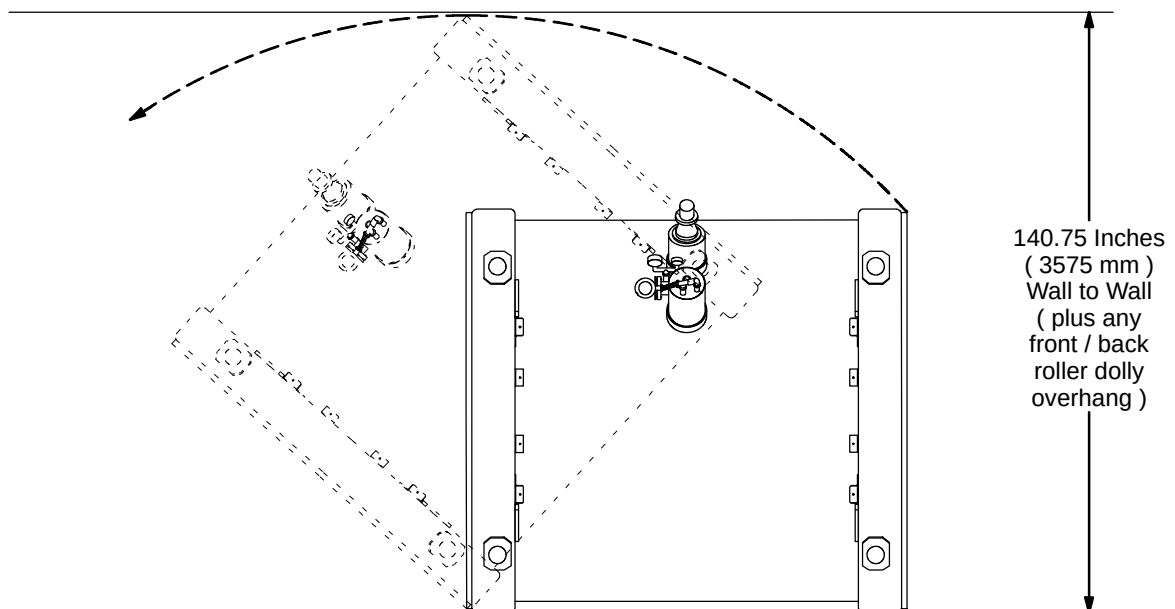
* Allow at least 1 inch (25 mm) additional clearance for protective packaging around Service Turret and Shield Cooler Cold-head. This packaging MUST be kept in place until magnet is in magnet room.

{ Exhaust Vent, show upright in drawing, ships rotated down.

{{ Minimum vertical clearance to commission / service magnet: 131.6 inches (3343 mm) with normal tools; 114.2 inches (2900 mm) with low ceiling height tools.

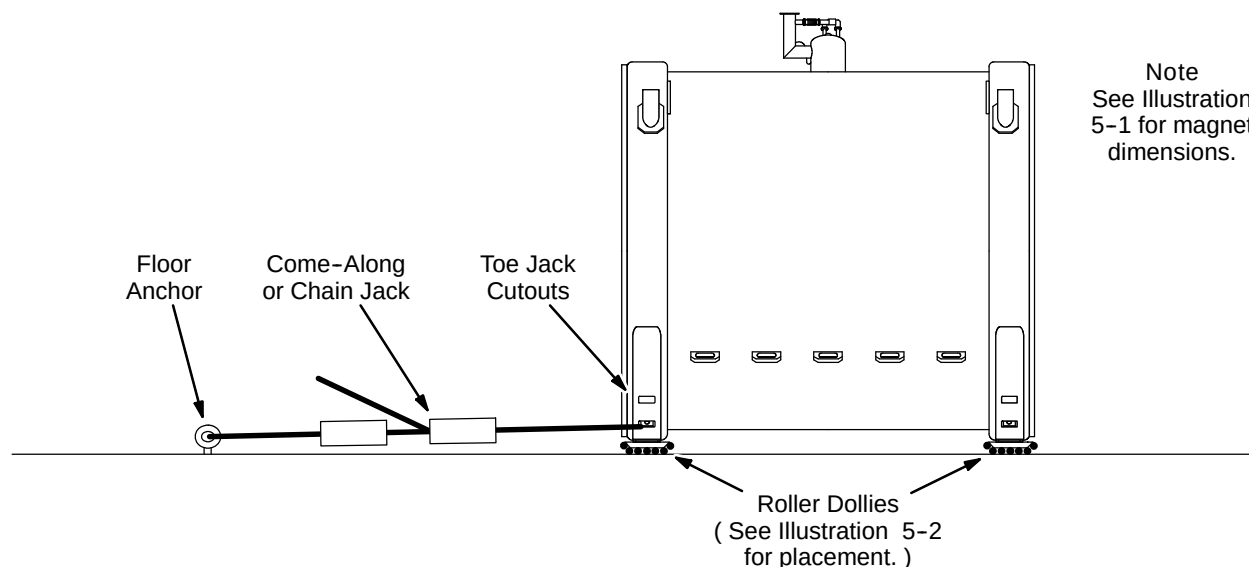
MAGNET ON ROLLER DOLLIES
ILLUSTRATION 5-1

5-1 INTRODUCTION (continued)

ROLLER DOLLY PLACEMENT
ILLUSTRATION 5-2MINIMUM HORIZONTAL CLEARANCE TO ROTATE MAGNET
ILLUSTRATION 5-3

5-1 INTRODUCTION (continued)

A motorized tow vehicle is recommended to help move the magnet. A come-along or chain jack such as shown in Illustration 5-4 can also be used. Attach cables, chains or straps to the magnet legs with shackles.



USING A COME-ALONG OR CHAIN JACK
ILLUSTRATION 5-4

5-2 MOVING THE MAGNET

1. Make sure the magnet anchors have been installed in conformance with Direction 2221775, SIGNA[®] 3.0T/94 & VH3 Pre-Installation, prior to moving magnet into magnet room.
2. Check the clearances along the route the magnet will move to the magnet room. Compare those clearances with Illustration 5-1.



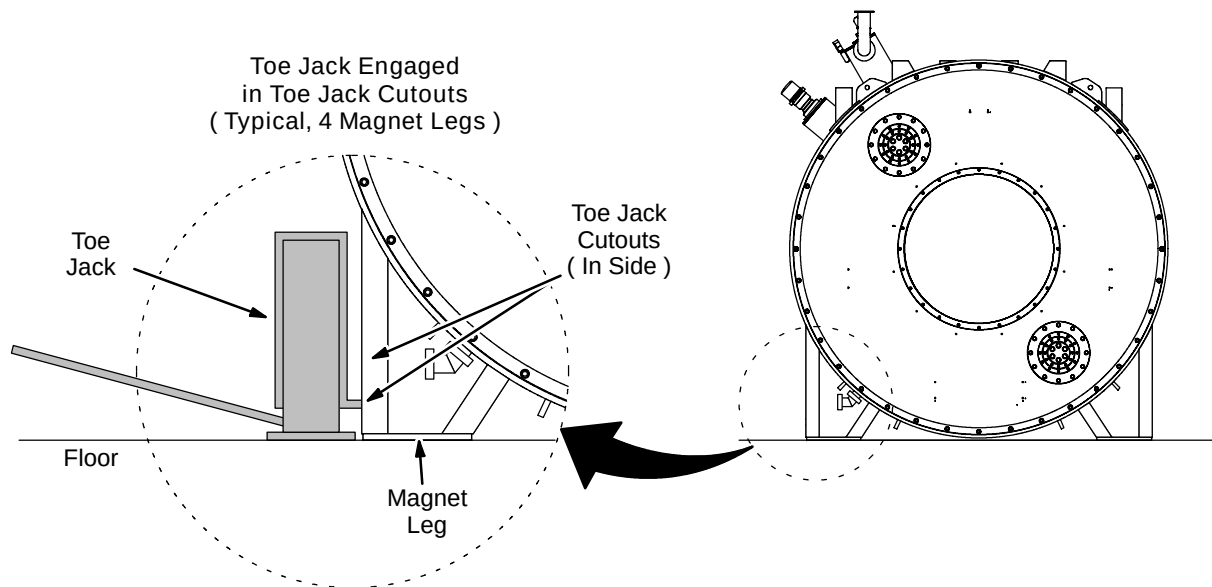
Keep the magnet level at all times. Uneven jacking of the magnet's corners could result in the magnet shifting on the jacks.

3. Smoothly and evenly raise the entire magnet using four side-lifting toe jacks as in Illustration 5-5.

Note

If all four toe jacks cannot be lifted simultaneously, lift two at a time. Raise both jacks on one end of the magnet 1 - 2 inches (25 - 50 mm), then raise both jacks on the other end until 1 - 2 inches (25 - 50 mm) higher than the first end. Continue in this manner until sufficient height is attained and the magnet is in a level and stable position.

5-2 MOVING THE MAGNET (continued)



JACKING THE MAGNET LEGS
ILLUSTRATION 5-5

4. Insert a roller dolly under each leg of the raised magnet. Lower the magnet fully onto the roller dollies.
5. Compare the dimensions of the magnet on the roller dollies with the clearances measured along the magnet delivery route. If the minimum overhead clearance will be less than 1 inch (25 mm), the magnet requires the removal of the Service Turret Top Plate and the Pressure Gauge Manifold.

Note

Adjustments to magnet / roller dolly height to meet clearance needs are best made before moving the magnet. If simple adjustments are not sufficient to meet those needs, call the MAC Team representative for removal of components.

6. Move the magnet to the magnet room. If using a motorized tow vehicle, attach cables, chains or straps to the magnet's legs with shackles.

Note

If there are turns in the delivery route, adjust the roller dollies to appropriate positions to negotiate turns.

Use shims to move a magnet on roller dollies over door thresholds or other inclines.

If the actual overhead clearance encountered while moving the magnet is less than 1 inch (25 mm), the magnet requires the removal of the Service Turret Top Plate and the Pressure Gauge Manifold. If further adjustments are required to meet overhead clearance needs, call the MAC Team representative for removal of components.

7. Identify the table end of the magnet before moving into magnet room.

5-2 MOVING THE MAGNET (continued)

8. Position the magnet over the anchor studs.

Note

It is recommended to protect anchor stud threads with tubing to prevent damage to the threads during magnet installation.



Keep the magnet level at all times. Uneven jacking of the magnet's corners could result in the magnet shifting on the jacks.

9. Jack the magnet up sufficiently at the four legs and remove the four roller dollies.
10. Slowly lower the magnet onto the magnet room floor: Simultaneously lower both toe jacks on one end of the magnet until that end is 1 - 2 inches (25 - 50 mm) lower than the opposite end. Then simultaneously lower both jacks on the other end 1 - 2 inches (25 - 50 mm). Repeat lowering the magnet end to end until all feet are on the floor, correctly located on the anchor studs.